MONISH KUMAR V R

PRINCIPAL ENGINEER AI - INDUSTRIAL AI s OPTIMIZATION

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Industrial AI Engineer with 4.5+ years architecting and owning production AI systems end-to-end — from model development through orchestration-layer design to fault-tolerant edge deployment across industrial environments. Experienced leading customer-facing AI workstreams from plant discovery and problem formulation through deployment, monitoring, and measurable business impact.

CORE EXPERTISE

Optimization s Operations Research:	MILP, constraint modeling, scenario analysis, production planning, charge mix optimization, PuLP, Pyomo
Machine Learning s Statistics:	Python, SQL, Scikit-learn, PyTorch, feature engineering, supervised learning, evaluation, data drift, automated retraining
Production s MLOps:	Docker, REST APIs, Linux, model monitoring, data-quality checks, AWS SageMaker and GCP Compute Engine exposure, edge-to-cloud deployment

EXPERIENCE

Principal Engineer - AI | NowPurchase, Kolkata — Sep 2021 - Present

- Built a charge-mix MILP optimizer using PuLP/Pyomo concepts to minimize melt cost under grade chemistry, raw-material availability, recovery, furnace capacity, and operational constraints; delivered 2-5% melt-cost savings.
- Developed a multi-stage production-planning optimizer across demand, furnace/line capacity, shift availability, and process constraints; improved OTIF and unlocked 5-8% effective production capacity.
- Productized quality and process ML models using spectrometer chemistry and operations data; improved accuracy from 75% to 95% and implemented data-drift monitoring, validation gates, and automated retraining triggers.
- Integrated machine-vision events with shift, operator, and process telemetry to generate cycle-time, tap-to-tap, pouring-duration, compliance, and bottleneck KPIs for supervisors and plant leadership.
- Designed a foundry-process ontology in MongoDB and Postgres and customer-specific AI agents that convert natural-language questions into governed queries over industrial data, enabling reusable decision-intelligence workflows.
- Owned the technical architecture for fleet-wide edge deployment reliability: designed staged, review-gated pipeline with deterministic artifact resolution and automatic rollback in network-constrained plants, including Dockerized services, monitoring, evidence capture, and human-in-the-loop validation; reduced test-to-production lead time from about 3 days to 1 day.
- Lead AI and data team members, review modeling and deployment approaches, and communicate technical trade-offs and business impact to customers, product teams, and leadership.

SELECTED INDUSTRIAL AI s OPTIMIZATION PROJECTS

Charge-Mix Prescriptive Optimizer *[MILP | PuLP | Pyomo | Manufacturing Economics]*

Formulated raw-material selection and quantity decisions as a constrained optimization problem with chemistry ranges, material recovery, inventory, furnace weight, and operational rules; generated explainable, cost-minimizing recommendations for melt planning.

Delivered 2-5% melt-cost reduction and created a reusable framework for customer-specific grades, materials, and operating constraints.

Multi-Stage Production Planning Optimizer *[Operations Research | Capacity Planning | OTIF]*

Optimized production allocation across stages and shifts using demand, throughput, furnace/line capacity, and sequencing constraints; improved on-time-in-full delivery and unlocked 5-8% effective capacity.

Quality s Process Intelligence Models *[Scikit-learn | Python | Process Data | Drift Monitoring]*

Built predictive models using spectrometer chemistry and operational features to improve quality/process decision support; raised model accuracy from 75% to 95%.

Implemented production monitoring, failure-slice analysis, data-drift detection, and automated retraining triggers to maintain reliability as raw materials and process behavior changed.

Melting s Pouring Process Intelligence *[Computer Vision | Time-Series Events | Process Telemetry]*

Converted continuous furnace and pouring video into structured heat-start, tapping, deslagging, heat-end, and pour-duration events, then correlated them with shift/operator data for cycle-time and bottleneck analysis.

Applied temporal filtering and state-machine validation to reduce event false triggers below 3% on production samples, with evidence snapshots and clips for auditability and root-cause review.

Industrial Knowledge s Decision Agent *[Process Ontology | Tool-Calling | LLM Workflows | MCP]*

Designed and deployed a Model Context Protocol server as the orchestration layer for agentic query workflows over industrial data — governed tool-calling, per-client OAuth-scoped context, reliable state management across seven integrated data sources (ERP, sensors, chemistry, production timelines), serving 100+ users; established context engineering (not model scale) as the architectural moat.

EDUCATION

M.Tech, Metallurgical s Materials Engineering - IIT Kharagpur — CGPA 8.63/10 | 2019-2021

B.E., Metallurgical Engineering - PSG College of Technology, Coimbatore — CGPA 8.93/10 | 2014-2018

RECOGNITION s PUBLICATIONS

World Foundry Organization - Top 10 Globally: Presented "Feedback-Driven Charge Optimization for Non-Digitized Foundries" at the World Foundry Congress, Hannover, Germany.

IIF Journal (Published): "ML + Game Theory to Reduce Energy Consumption in Induction Furnace Melting" - applied ML and optimization research for constrained industrial environments.

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